Bash Script Tasks

Create a bash script to **check if a directory is available or not**.

```
#!/bin/bash

if [ -e /rohith/scripts ]; then
  echo "File exists"
else
  echo "File does not exist"
fi
~
```

```
[root@ip-172-31-5-159 scripts]# vi check.bash
[root@ip-172-31-5-159 scripts]# chmod 755 check.bash
[root@ip-172-31-5-159 scripts]# ./check.bash
File exists
```

Create a bash script to **create multiple files**.

```
[root@ip-172-31-5-159 scripts]# vi file.bash
[root@ip-172-31-5-159 scripts]# chmod 755 file.bash
[root@ip-172-31-5-159 scripts]# ./file.bash
[root@ip-172-31-5-159 scripts]# ls
Defining_Tasks.bash      check.bash     forloop2.bash    ifthen4.bash
Defining_variables.bash  commands.bash  forloop3.bash    input.bash
Output_to_screen.bash    file.bash      forloop4.bash    no_of_secs.bash
a                        file1.txt      forloop5.bash    no_of_times.bash
admin.bash               file2.txt      forloop6.bash    perform_task.bash
automate.bash            file3.txt      helloworld.bash  run_command.bash
b                        file4.txt      ifthen1.bash     state_of_backup.bash
c                        file5.txt      ifthen2.bash     task_performed.bash
case_scripts.bash        forloop1.bash  ifthen3.bash     variables.bash
```

```
#!/bin/bash

for i in {1..5}
do
  touch "file$i.txt"
done
~
~
```

Create a bash script to **take a backup of a directory**.

```bash
#!/bin/bash

# Source directory (the directory you want to backup)
SOURCE="/rohith/scripts"

# Destination directory (where backup will be stored)
DEST="/rohith/backup"

# Create destination directory if it doesn't exist
mkdir -p $DEST

# Create backup file name with date
DATE=$(date +%Y-%m-%d_%H-%M-%S)

# Backup file name
BACKUP_FILE="$DEST/backup_$DATE.tar.gz"

# Create backup using tar
tar -czvf $BACKUP_FILE $SOURCE

# Print success message
echo "Backup completed successfully!"
echo "Backup file: $BACKUP_FILE"
~
~
```

```
[root@ip-172-31-5-159 ~]# vi file.bash
[root@ip-172-31-5-159 ~]# cd /rohith/scripts
[root@ip-172-31-5-159 scripts]# vi
[root@ip-172-31-5-159 scripts]# ls
Defining_Tasks.bash      check.bash     forloop2.bash    ifthen4.bash
Defining_variables.bash  commands.bash  forloop3.bash    input.bash
Output_to_screen.bash    file.bash      forloop4.bash    no_of_secs.bash
a                        file1.txt      forloop5.bash    no_of_times.bash
admin.bash               file2.txt      forloop6.bash    perform_task.bash
automate.bash            file3.txt      helloworld.bash  run_command.bash
b                        file4.txt      ifthen1.bash     state_of_backup.bash
c                        file5.txt      ifthen2.bash     task_performed.bash
case_scripts.bash        forloop1.bash  ifthen3.bash     variables.bash
[root@ip-172-31-5-159 scripts]# pwd
/rohith/scripts
[root@ip-172-31-5-159 scripts]# cd..
-bash: cd..: command not found
[root@ip-172-31-5-159 scripts]# cd ..
[root@ip-172-31-5-159 rohith]# mkdir backup
[root@ip-172-31-5-159 rohith]# cd scripts
[root@ip-172-31-5-159 scripts]# vi backup.sh
[root@ip-172-31-5-159 scripts]# chmod 755 backup.sh
[root@ip-172-31-5-159 scripts]# ./backup.sh
tar: Removing leading `/' from member names
/rohith/scripts/
/rohith/scripts/Defining_Tasks.bash
/rohith/scripts/Defining_variables.bash
/rohith/scripts/Output_to_screen.bash
/rohith/scripts/automate.bash
/rohith/scripts/case_scripts.bash
/rohith/scripts/forloop1.bash
/rohith/scripts/forloop2.bash
/rohith/scripts/forloop3.bash
/rohith/scripts/forloop4.bash
/rohith/scripts/forloop5.bash
/rohith/scripts/forloop6.bash
/rohith/scripts/ifthen1.bash
/rohith/scripts/ifthen2.bash
/rohith/scripts/ifthen3.bash
/rohith/scripts/ifthen4.bash
/rohith/scripts/no_of_secs.bash
/rohith/scripts/no_of_times.bash
/rohith/scripts/perform_task.bash
/rohith/scripts/run_command.bash
/rohith/scripts/state_of_backup.bash
/rohith/scripts/task_performed.bash
/rohith/scripts/a
/rohith/scripts/b
/rohith/scripts/c
/rohith/scripts/admin.bash
/rohith/scripts/commands.bash
/rohith/scripts/helloworld.bash
/rohith/scripts/input.bash
/rohith/scripts/variables.bash
/rohith/scripts/check.bash
/rohith/scripts/file.bash
/rohith/scripts/file1.txt
/rohith/scripts/file2.txt
/rohith/scripts/file3.txt
/rohith/scripts/file4.txt
/rohith/scripts/file5.txt
/rohith/scripts/backup.sh
Backup completed successfully!
```

```
/rohith/scripts/backup.sh
Backup completed successfully!
Backup file: /rohith/backup/backup_2026-04-03_11-48-37.tar.gz
[root@ip-172-31-5-159 scripts]# ls
Defining_Tasks.bash      check.bash     forloop3.bash    no_of_secs.bash
Defining_variables.bash  commands.bash  forloop4.bash    no_of_times.bash
Output_to_screen.bash    file.bash      forloop5.bash    perform_task.bash
a                        file1.txt      forloop6.bash    run_command.bash
admin.bash               file2.txt      helloworld.bash  state_of_backup.bash
automate.bash            file3.txt      ifthen1.bash     task_performed.bash
b                        file4.txt      ifthen2.bash     variables.bash
backup.sh                file5.txt      ifthen3.bash
c                        forloop1.bash  ifthen4.bash
case_scripts.bash        forloop2.bash  input.bash
[root@ip-172-31-5-159 scripts]# cd ..
[root@ip-172-31-5-159 rohith]# ls
backup  scripts
[root@ip-172-31-5-159 rohith]# cd backup
[root@ip-172-31-5-159 backup]# ls
backup_2026-04-03_11-48-37.tar.gz
[root@ip-172-31-5-159 backup]# tar -tzvf backup_2026-04-03_11-48-37.tar.gz
drwxr-xr-x root/root         0 2026-04-03 11:47 rohith/scripts/
-rwxr-xr-x root/root        83 2026-04-02 11:50 rohith/scripts/Defining_Tasks.ba
sh
-rwxr-xr-x root/root       155 2026-04-02 12:01 rohith/scripts/Defining_variable
s.bash
-rwxr-xr-x root/root        56 2026-04-02 11:47 rohith/scripts/Output_to_screen.
bash
-rwxr-xr-x root/root       679 2026-04-02 12:58 rohith/scripts/automate.bash
-rwxr-xr-x root/root       307 2026-04-02 12:44 rohith/scripts/case_scripts.bash
-rwxr-xr-x root/root        63 2026-04-02 13:13 rohith/scripts/forloop1.bash
-rwxr-xr-x root/root        65 2026-04-02 13:15 rohith/scripts/forloop2.bash
-rwxr-xr-x root/root        45 2026-04-02 13:16 rohith/scripts/forloop3.bash
-rwxr-xr-x root/root        42 2026-04-02 13:21 rohith/scripts/forloop4.bash
-rwxr-xr-x root/root        86 2026-04-02 13:23 rohith/scripts/forloop5.bash
-rwxr-xr-x root/root       112 2026-04-02 13:25 rohith/scripts/forloop6.bash
-rwxrwxrwx root/root        98 2026-04-02 13:30 rohith/scripts/ifthen1.bash
-rwxrwxrwx root/root       110 2026-04-02 13:32 rohith/scripts/ifthen2.bash
-rwxrwxrwx root/root       115 2026-04-02 13:34 rohith/scripts/ifthen3.bash
-rwxrwxrwx root/root       267 2026-04-02 13:35 rohith/scripts/ifthen4.bash
-rwxr-xr-x root/root       209 2026-04-02 13:10 rohith/scripts/no_of_secs.bash
-rwxr-xr-x root/root        77 2026-04-02 13:08 rohith/scripts/no_of_times.bash
-rwxr-xr-x root/root       223 2026-04-02 12:36 rohith/scripts/perform_task.bash
-rwxr-xr-x root/root       154 2026-04-02 12:18 rohith/scripts/run_command.bash
-rwxr-xr-x root/root       149 2026-04-02 13:01 rohith/scripts/state_of_backup.b
ash
-rwxr-xr-x root/root       134 2026-04-02 12:15 rohith/scripts/task_performed.ba
sh
-rw-r--r-- root/root         0 2026-04-02 10:41 rohith/scripts/a
-rw-r--r-- root/root         0 2026-04-02 10:41 rohith/scripts/b
-rw-r--r-- root/root         0 2026-04-02 10:41 rohith/scripts/c
-rwxr-xr-x root/root        58 2026-04-02 10:49 rohith/scripts/admin.bash
-rwxr-xr-x root/root        47 2026-04-02 10:41 rohith/scripts/commands.bash
-rwxr-xr-x root/root       104 2026-04-02 10:19 rohith/scripts/helloworld.bash
-rwxr-xr-x root/root       108 2026-04-02 11:00 rohith/scripts/input.bash
-rwxr-xr-x root/root       118 2026-04-02 10:56 rohith/scripts/variables.bash
-rwxr-xr-x root/root       103 2026-04-03 10:56 rohith/scripts/check.bash
-rwxr-xr-x root/root        58 2026-04-03 11:05 rohith/scripts/file.bash
-rw-r--r-- root/root         0 2026-04-03 11:06 rohith/scripts/file1.txt
-rw-r--r-- root/root         0 2026-04-03 11:06 rohith/scripts/file2.txt
-rw-r--r-- root/root         0 2026-04-03 11:06 rohith/scripts/file3.txt
-rw-r--r-- root/root         0 2026-04-03 11:06 rohith/scripts/file4.txt
-rw-r--r-- root/root         0 2026-04-03 11:06 rohith/scripts/file5.txt
-rwxr-xr-x root/root       519 2026-04-03 11:47 rohith/scripts/backup.sh
[root@ip-172-31-5-159 backup]#
```

Create a bash script to **install Nginx on an EC2 server**.

```
#!/bin/bash

echo "Updating system packages..."
sudo yum update -y

echo "Installing Nginx..."
sudo yum install nginx -y

echo "Starting Nginx service..."
sudo systemctl start nginx

echo "Enabling Nginx on boot..."
sudo systemctl enable nginx

echo "Checking Nginx status..."
sudo systemctl status nginx
```

```
Installed:
  gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
  libunwind-1.4.0-5.amzn2023.0.3.x86_64
  nginx-1:1.28.2-1.amzn2023.0.1.x86_64
  nginx-core-1:1.28.2-1.amzn2023.0.1.x86_64
  nginx-filesystem-1:1.28.2-1.amzn2023.0.1.noarch
  nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
Starting Nginx service...
Enabling Nginx on boot...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr
/lib/systemd/system/nginx.service.
Checking Nginx status...
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: di>
     Active: active (running) since Fri 2026-04-03 12:14:53 UTC; 378ms ago
   Main PID: 3765 (nginx)
      Tasks: 3 (limit: 1043)
     Memory: 3.3M
        CPU: 60ms
     CGroup: /system.slice/nginx.service
             ├─3765 "nginx: master process /usr/sbin/nginx"
             ├─3766 "nginx: worker process"
             └─3767 "nginx: worker process"

Apr 03 12:14:53 ip-172-31-5-159.ec2.internal systemd[1]: Starting nginx.service>
Apr 03 12:14:53 ip-172-31-5-159.ec2.internal nginx[3763]: nginx: the configurat>
Apr 03 12:14:53 ip-172-31-5-159.ec2.internal nginx[3763]: nginx: configuration >
Apr 03 12:14:53 ip-172-31-5-159.ec2.internal systemd[1]: Started nginx.service >
[root@ip-172-31-5-159 scripts]#
[root@ip-172-31-5-159 scripts]# vi install_nginx.bash
[root@ip-172-31-5-159 scripts]#
```

Create a bash script to **install Apache Tomcat on an EC2 server**.

```
#!/bin/bash
# Installing Apache Tomcat
cd /opt/
wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.20/bin/apache-tomcat-11.0.20.tar.gz

gar xvf apache-tomcat-11.0.20.tar.gz

yum install java-17-amazon-corretto -y

cd apache-tomcat-11.0.20

cd bin

sh startup.sh



echo Apache Tomcat is installed
~
~
~
```

```
[root@ip-172-31-5-159 scripts]#
[root@ip-172-31-5-159 scripts]# vi install_nginx.bash
[root@ip-172-31-5-159 scripts]# vi install_tomcat.bash
[root@ip-172-31-5-159 scripts]# chmod 755 install_tomcat.bash
[root@ip-172-31-5-159 scripts]# ./install_tomcat.bash
--2026-04-03 12:22:27--  https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.20/bin/apache-tomcat-11.0.20.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14353103 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-11.0.20.tar.gz'

apache-tomcat-11.0.20.tar.gz                      100%[===================================================================================================>]  13.69M  --.-KB/s    in 0.07s

2026-04-03 12:22:27 (193 MB/s) - 'apache-tomcat-11.0.20.tar.gz' saved [14353103/14353103]

./install_tomcat.bash: line 6: gar: command not found
Last metadata expiration check: 5:41:32 ago on Fri Apr  3 06:40:55 2026.
Dependencies resolved.
==================================================================================================================================================
 Package                                                Architecture                    Version                                    Repository                                   Size
==================================================================================================================================================
Installing:
 java-17-amazon-corretto                                x86_64                          1:17.0.18+9-1.amzn2023.1                   amazonlinux                                 219 k
Installing dependencies:
 alsa-lib                                               x86_64                          1.2.7.2-1.amzn2023.0.3                     amazonlinux                                 503 k
 cairo                                                  x86_64                          1.18.0-4.amzn2023.0.3                      amazonlinux                                 717 k
 dejavu-sans-fonts                                      noarch                          2.37-16.amzn2023.0.2                       amazonlinux                                 1.3 M
 dejavu-sans-mono-fonts                                 noarch                          2.37-16.amzn2023.0.2                       amazonlinux                                 467 k
 dejavu-serif-fonts                                     noarch                          2.37-16.amzn2023.0.2                       amazonlinux                                 1.0 M
 fontconfig                                             x86_64                          2.13.94-2.amzn2023.0.2                     amazonlinux                                 273 k
 fonts-filesystem                                       noarch                          1:2.0.5-12.amzn2023.0.2                    amazonlinux                                 9.5 k
 freetype                                               x86_64                          2.13.2-5.amzn2023.0.2                      amazonlinux                                 422 k
 giflib                                                 x86_64                          5.2.1-9.amzn2023.0.2                       amazonlinux                                  48 k
 google-noto-fonts-common                               noarch                          20240401-1.amzn2023.0.2                    amazonlinux                                  17 k
 google-noto-sans-vf-fonts                              noarch                          20240401-1.amzn2023.0.2                    amazonlinux                                 593 k
 graphite2                                              x86_64                          1.3.14-7.amzn2023.0.2                      amazonlinux                                  97 k
 harfbuzz                                               x86_64                          7.0.0-2.amzn2023.0.2                       amazonlinux                                 873 k
 java-17-amazon-corretto-headless                       x86_64                          1:17.0.18+9-1.amzn2023.1                   amazonlinux                                  91 M
 javapackages-filesystem                                noarch                          6.0.0-7.amzn2023.0.6                       amazonlinux                                  12 k
 langpacks-core-font-en                                 noarch                          3.0-21.amzn2023.0.4                        amazonlinux                                  10 k
 libICE                                                 x86_64                          1.1.1-3.amzn2023.0.1                       amazonlinux                                  76 k
 libSM                                                  x86_64                          1.2.4-3.amzn2023.0.1                       amazonlinux                                  45 k
 libX11                                                 x86_64                          1.8.10-2.amzn2023.0.1                      amazonlinux                                 659 k
 libX11-common                                          noarch                          1.8.10-2.amzn2023.0.1                      amazonlinux                                 147 k
 libXau                                                 x86_64                          1.0.11-6.amzn2023.0.1                      amazonlinux                                  33 k
 libXext                                                x86_64                          1.3.6-1.amzn2023.0.1                       amazonlinux                                  42 k
 libXi                                                  x86_64                          1.8.2-1.amzn2023.0.1                       amazonlinux                                  42 k
 libXinerama                                            x86_64                          1.1.5-6.amzn2023.0.1                       amazonlinux                                  16 k
 libXrandr                                              x86_64                          1.5.4-3.amzn2023.0.1                       amazonlinux                                  29 k
 libXrender                                             x86_64                          0.9.11-6.amzn2023.0.1                      amazonlinux                                  29 k
 libXt                                                  x86_64                          1.3.0-3.amzn2023.0.1                       amazonlinux                                 183 k
 libXtst                                                x86_64                          1.2.5-1.amzn2023.0.1                       amazonlinux                                  22 k
 libjpeg-turbo                                          x86_64                          2.1.4-2.amzn2023.0.5                       amazonlinux                                 190 k
 libpng                                                 x86_64                          2:1.6.37-10.amzn2023.0.11                  amazonlinux                                 128 k
 libxcb                                                 x86_64                          1.17.0-1.amzn2023.0.1                      amazonlinux                                 235 k
 pixman                                                 x86_64                          0.43.4-1.amzn2023.0.4                      amazonlinux                                 296 k
 xml-common                                             noarch                          0.6.3-56.amzn2023.0.2                      amazonlinux                                  32 k

Transaction Summary
==================================================================================================================================================
```

Create a bash script to **check if the Nginx service is running**, if not running then script should start the service.

root@ip-172-31-5-159:/rohith/scripts

```
#!/bin/bash
#nginx start

if systemctl status nginx; then
    echo "Nginx is running"
else
    echo "Nginx not running. Starting..."
     systemctl start nginx
fi
~
~
~
~
```

```
[root@ip-172-31-5-159 scripts]# chmod 755 check_nginx.bash
[root@ip-172-31-5-159 scripts]# ./check_nginx.bash
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Fri 2026-04-03 12:14:53 UTC; 12min ago
   Main PID: 3765 (nginx)
      Tasks: 3 (limit: 1043)
     Memory: 3.2M
        CPU: 60ms
     CGroup: /system.slice/nginx.service
             ├─3765 "nginx: master process /usr/sbin/nginx"
             ├─3766 "nginx: worker process"
             └─3767 "nginx: worker process"

Apr 03 12:14:53 ip-172-31-5-159.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Apr 03 12:14:53 ip-172-31-5-159.ec2.internal nginx[3763]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Apr 03 12:14:53 ip-172-31-5-159.ec2.internal nginx[3763]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Apr 03 12:14:53 ip-172-31-5-159.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
./check_nginx.bash: line 5:     echo: command not found
[root@ip-172-31-5-159 scripts]# |
```

Create a bash script for a **calculator**.

root@ip-172-31-5-159:/rohith/scripts

```
#!/bin/bash

echo "Enter first number:"
read num1

echo "Enter second number:"
read num2

echo "Choose operation: +  -  *  /"
read op

if [ "$op" == "+" ]; then
    echo "Result: $(($num1 + $num2))"

elif [ "$op" == "-" ]; then
    echo "Result: $(($num1 - $num2))"

elif [ "$op" == "*" ]; then
    echo "Result: $(($num1 * $num2))"

elif [ "$op" == "/" ]; then
    echo "Result: $(($num1 / $num2))"

else
    echo "Invalid operation"
fi
~
~
~
```

```
[root@ip-172-31-5-159 scripts]# vi check_nginx.bash
[root@ip-172-31-5-159 scripts]# vi calculator.bash
[root@ip-172-31-5-159 scripts]# chmod 755 calculator.bash
[root@ip-172-31-5-159 scripts]# ./calculator.bash
Enter first number:
4
Enter second number:
5
Choose operation: +  -  *  /
1
Invalid operation
[root@ip-172-31-5-159 scripts]# ./calculator.bash
Enter first number:
4
Enter second number:
5
Choose operation: +  -  *  /
+
Result: 9
[root@ip-172-31-5-159 scripts]#
```

Create a bash script to **check if a directory exists**, if not then create a directory.

```
[root@ip-172-31-5-159 scripts]# cd /home/ec2-user/mydir
[root@ip-172-31-5-159 mydir]# cd /home/ec2-user
[root@ip-172-31-5-159 ec2-user]# ll
total 0
drwxr-xr-x. 2 root root 6 Apr  3 12:35 mydir
[root@ip-172-31-5-159 ec2-user]#
```

:

root@ip-172-31-5-159:/rohith/scripts

```
#!/bin/bash

DIR="/home/ec2-user/mydir"

if [ -d "$DIR" ]; then
    echo "Directory already exists"
else
    echo "Directory not found. Creating directory..."
    mkdir -p "$DIR"
    echo "Directory created successfully"
fi
~
~
~
~
```

1. Create a bash script to **delete the last 3 lines of a file**.
2. Bash script to monitor cpu and if it is more than 80% then send email notification.
3. Bash script to monitor disk space and if it is more than 80% then send email notification.
4. Bash script to monitor memory and if it is more than 80% then send email notification.

Exercise: Crontab Entries

April 5th Midnight

5th of Every November, January, June **if it is a Thursday**

1. At **05 and 27th minutes** of **9, 10, 11** hours every day

34 minutes of **9th hour** on **15th August**
Every midnight

1. Every Weekend (**Saturday night 11:59 PM**)
2. After every reboot

Notes
All the scripts need to be executed on **Amazon Linux EC2**.
Document all the tasks and also push the tasks to a **GitHub repository** (explore this for now).

1. Share the **document** and **GitHub Repo URL** once done for evaluation.